

④3 Дифференцирование системы
дифференциальных уравнений

$Z = f(x_1, x_2, x_3, \dots, x_n)$ дифференцируема в
точке $(x_1^0, \dots, x_n^0)$

$$x_1 = \varphi_1(t_1, \dots, t_k)$$

$$x_2 = \varphi_2(t_1, \dots, t_k)$$

... — — — — —

$$x_n = \varphi_n(t_1, \dots, t_k)$$

дифференцируема в точке $N_0(t_1^0, \dots, t_k^0)$

Тогда данное $z = f(y_1, t_1, \dots, t_k), \dots, y_n(t_1, \dots, t_k)$

Основная теорема

- 1) $z = f(x_1, \dots, x_n)$ непрерывно в точке $N_0(x_1^0, \dots, x_n^0)$
- 2) $\forall i = \overline{1, n}$; $x_i = y_i(t_1, \dots, t_k)$ непрерывно в точке $N_0(t_1^0, \dots, t_k^0)$

Тогда данное $z = f(x_1(t_1, \dots, t_k), \dots, x_n(t_1, \dots, t_k))$

непрерывно по $t_1, \dots, t_k$ и

$$\frac{\partial Z}{\partial t_j} = \frac{\partial F}{\partial x_1} \frac{\partial x_1}{\partial t_j} + \frac{\partial F}{\partial x_2} \frac{\partial x_2}{\partial t_j} + \dots + \frac{\partial F}{\partial x_n} \frac{\partial x_n}{\partial t_j}, \quad j = \overline{1, n}$$

Доказательство: Пусть $Z = F(x_1(t_1, \dots, t_n), \dots, x_n(t_1, \dots, t_n)) =$
 $= F(t_1, \dots, t_n)$ рассмотрим приращение:

$$\therefore F(t_1 + \Delta t_1, \dots, t_n + \Delta t_n) - F(t_1, \dots, t_n) =$$

$$= L_1^i \Delta t_1 + \dots + L_n^i \Delta t_n + o(\rho), \quad \rho = \sqrt{\Delta t_1^2 + \Delta t_2^2 + \dots + \Delta t_n^2}$$

Введем же $N_0(t_1^0, \dots, t_n^0)$: $\Delta x_i = x_i(N) - x_i(N_0) =$

$$= A_1^i \Delta t_1 + \dots + A_n^i \Delta t_n + o(\rho) = \frac{\partial x_i}{\partial t_1} \Delta t_1 + \dots + \frac{\partial x_i}{\partial t_n} \Delta t_n + o(\rho)$$

$\forall i = \overline{1, n}$

$N_0 \approx u_0, N_1 \approx u_1$

$$\Delta z = f(x_1, \dots, x_n) - f(x_1^0, \dots, x_n^0) = f(u) - f(u_0) = \beta_1 \Delta x_1 + \dots + \beta_n \Delta x_n$$

$$+ o(r) = \frac{\partial z}{\partial x_1} \Delta x_1 + \dots + \frac{\partial z}{\partial x_n} \Delta x_n + o(r) \quad (\text{E}) \quad \left(r = \sqrt{\Delta x_1^2 + \dots + \Delta x_n^2} \right) \quad (\equiv)$$

$$(\text{E}) \quad \frac{\partial z}{\partial x_1} \left(\frac{\partial x_1}{\partial t_1} \Delta t_1 + \dots + \frac{\partial x_1}{\partial t_k} \Delta t_k + o(r) \right) + \dots +$$

$$+ \frac{\partial z}{\partial x_n} \left(\frac{\partial x_n}{\partial t_1} \Delta t_1 + \dots + \frac{\partial x_n}{\partial t_k} \Delta t_k + o(r) \right) + o(r) =$$

$$= \left(\frac{\partial z}{\partial x_1} \frac{\partial x_1}{\partial t_1} + \frac{\partial z}{\partial x_2} \frac{\partial x_2}{\partial t_1} + \dots + \frac{\partial z}{\partial x_n} \frac{\partial x_n}{\partial t_1} \right) \Delta t_1 + \dots +$$

$$+ \left(\frac{\partial z}{\partial x_1} \frac{\partial x_1}{\partial t_k} + \dots + \frac{\partial z}{\partial x_n} \frac{\partial x_n}{\partial t_k} \right) \Delta t_k + \left(\frac{\partial z}{\partial x_1} + \frac{\partial z}{\partial x_2} + \dots + \frac{\partial z}{\partial x_n} \right) o(\rho) +$$

$$+ o(\tau) = L_1 \Delta t_1 + L_2 \Delta t_2 + \dots + L_k \Delta t_k + o(\rho) + o(\tau)$$

Помним, что $o(\tau) = o(\rho) : \frac{o(\tau)}{\rho} = \frac{o(\tau)}{\tau} \cdot \frac{\tau}{\rho}$

$$o(\tau) = \alpha_1 (\Delta x_1) \Delta x_1 + \dots + \alpha_n (\Delta x_n) \Delta x_n = \alpha_1 \left(\frac{\partial x_1}{\partial t_1} \Delta t_1 + \dots + \frac{\partial x_1}{\partial t_k} \Delta t_k + \right.$$

$$\left. + o(\rho) \right) + \dots + \alpha_n \left(\frac{\partial x_n}{\partial t_1} \Delta t_1 + \dots + \frac{\partial x_n}{\partial t_k} \Delta t_k + o(\rho) \right) =$$

$$= \left(\alpha_1 \frac{\partial x_1}{\partial t_1} + \dots + \alpha_n \frac{\partial x_n}{\partial t_1} \right) \Delta t_1 + \dots + \left(\alpha_1 \frac{\partial x_1}{\partial t_k} + \dots + \alpha_n \frac{\partial x_n}{\partial t_k} \right) \Delta t_k$$

$$+ (\alpha_1 + \dots + \alpha_n) (o(\rho)) = o(\rho)$$

В криволинейных координатах

1) Пусть $z = z(x(u, v), y(u, v)) = F(u, v)$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

2) Пусть $z = z(x(t), y(t))$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

$$3) \quad z = z(x, u, v)$$

$$\frac{\partial z}{\partial u} = \frac{dz}{dx} \cdot \frac{\partial x}{\partial u}$$

$$\frac{\partial z}{\partial v} = \frac{dz}{dx} \cdot \frac{\partial x}{\partial v}$$